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Project Title: College Admissions & Institutional Performance Analysis

Problem Statement:

I am analyzing a dataset of 777 US colleges to understand what institutional factors—such as application acceptance rates, faculty qualifications, expenditure per student, and student-to-faculty ratios—influence graduation rates. This analysis can help educational policymakers and prospective students identify which college characteristics are most strongly associated with student success and institutional quality.

Dataset Description:

- **Source:** ISLR R package / Kaggle (US College Dataset)
- **Rows:** 777
- **Columns:** 19
- **Column Descriptions:**
 - **College** – Name of the college
 - **Private** – Whether the college is private (Yes/No) — *Categorical*
 - **Apps** – Number of applications received
 - **Accept** – Number of applications accepted
 - **Enroll** – Number of new students enrolled
 - **Top10perc** – % of new students from top 10% of high school class
 - **Top25perc** – % of new students from top 25% of high school class
 - **F.Undergrad** – Number of full-time undergraduates
 - **P.Undergrad** – Number of part-time undergraduates
 - **Outstate** – Out-of-state tuition (
 - **Books** – Estimated book costs (
 - **PhD** – % of faculty with PhD
 - **Terminal** – % of faculty with terminal degree
 - **S.F.Ratio** – Student-to-faculty ratio
 - **perc.alumni** – % of alumni who donate

- Expend – Instructional expenditure per student (\$)
- Grad.Rate – Graduation rate (%)

Section 1: Import Libraries

```
[ ]
# All library imports in one cell
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import mean_squared_error, r2_score

# Display settings
pd.set_option('display.max_columns', None)
plt.rcParams['figure.figsize'] = (10, 6)
sns.set_style('whitegrid')

print('All libraries imported successfully!')
```

... All libraries imported successfully!

Section 2: Load Dataset

```
[ ]
# Load dataset – upload College.csv first using Files panel in Colab
df = pd.read_csv('College.csv', index_col=0)
df.index.name = 'College'
df.reset_index(inplace=True)

# Display first 5 rows
print('First 5 rows:')
df.head()
```

... First 5 rows:

	College	Private	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Undergrad	Outstate	Room.Board	Books	Personal	PhD	Terminal	S.F.Ratio	perc.alumni	Expend	Grad
0	Abilene Christian University	Yes	1660	1232	721	23	52	2885	537	7440	3300	450	2200	70	78	18.1	12	7041	
1	Adelphi University	Yes	2186	1924	512	16	29	2683	1227	12280	6450	750	1500	29	30	12.2	16	10527	
2	Adrian College	Yes	1428	1097	336	22	50	1036	99	11250	3750	400	1165	53	66	12.9	30	8735	
3	Agnes Scott College	Yes	417	349	137	60	89	510	63	12960	5450	450	875	92	97	7.7	37	19016	
4	Alaska Pacific University	Yes	193	146	55	16	44	249	869	7560	4120	800	1500	76	72	11.9	2	10922	



```
# Shape and column data types
print(f'Shape: {df.shape}')
print(f'\nColumns and Dtypes:')
print(df.dtypes)
```

... Shape: (777, 19)

Columns and Dtypes:

College	object
Private	object
Apps	int64
Accept	int64
Enroll	int64
Top10perc	int64
Top25perc	int64
F.Undergrad	int64
P.Undergrad	int64
Outstate	int64
Room.Board	int64
Books	int64
Personal	int64
PhD	int64
Terminal	int64
S.F.Ratio	float64
perc.alumni	int64
Expend	int64
Grad.Rate	int64
dtype:	object

Section 3: Data Cleaning & Statistics

[1]

```
# Check for missing values
print('Missing values per column:')
print(df.isnull().sum())
print(f'\nTotal missing values: {df.isnull().sum().sum()}')
```

```
... Missing values per column:
College      0
Private      0
Apps         0
Accept       0
Enroll       0
Top10perc    0
Top25perc    0
F.Undergrad  0
P.Undergrad  0
Outstate     0
Room.Board   0
Books        0
Personal     0
PhD          0
Terminal     0
S.F.Ratio    0
perc.alumni  0
Expend       0
Grad.Rate    0
dtype: int64

Total missing values: 0
```

```
# Check for duplicate rows
duplicates = df.duplicated().sum()
print(f'Duplicate rows: {duplicates}')
if duplicates > 0:
    df.drop_duplicates(inplace=True)
    print('Duplicates removed.')
else:
    print('No duplicates found. No action needed.')
```

```
... Duplicate rows: 0
No duplicates found. No action needed.
```

Data Cleaning Summary:

The dataset had no missing values and no duplicate rows, so no imputation or removal was necessary. The dataset is clean and ready for analysis.

```
# Descriptive Statistics for all numeric columns
df.describe()
```

	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Undergrad	Outstate	Room.Board	Books	Personal	PhD	Terminal	S.F.Ratio
count	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000
mean	3001.638353	2018.804376	779.972973	27.558559	55.796654	3699.907336	855.298584	10440.669241	4357.526384	549.380952	1340.642214	72.660232	79.702703	14.089704
std	3870.201484	2451.113971	929.176190	17.640364	19.804778	4850.420531	1522.431887	4023.016484	1096.696416	165.105360	677.071454	16.328155	14.722359	3.958349
min	81.000000	72.000000	35.000000	1.000000	9.000000	139.000000	1.000000	2340.000000	1780.000000	96.000000	250.000000	8.000000	24.000000	2.500000
25%	776.000000	604.000000	242.000000	15.000000	41.000000	992.000000	95.000000	7320.000000	3597.000000	470.000000	850.000000	62.000000	71.000000	11.500000
50%	1558.000000	1110.000000	434.000000	23.000000	54.000000	1707.000000	353.000000	9990.000000	4200.000000	500.000000	1200.000000	75.000000	82.000000	13.600000
75%	3624.000000	2424.000000	902.000000	35.000000	69.000000	4005.000000	967.000000	12925.000000	5050.000000	600.000000	1700.000000	85.000000	92.000000	16.500000
max	48094.000000	26330.000000	6392.000000	96.000000	100.000000	31643.000000	21836.000000	21700.000000	8124.000000	2340.000000	6800.000000	103.000000	100.000000	39.800000

```
# Detailed statistics for 2 key numeric columns: Grad.Rate and Outstate
cols = ['Grad.Rate', 'Outstate']
```

```
for col in cols:
    data = df[col]
    print(f'--- {col} ---')
    print(f'   Mean      : {data.mean():.2f}')
    print(f'   Median    : {data.median():.2f}')
    print(f'   Mode      : {data.mode()[0]:.2f}')
    print(f'   Std Dev   : {data.std():.2f}')
    print(f'   Variance  : {data.var():.2f}')
    print(f'   Range     : {data.max() - data.min():.2f}')
    print(f'   Mid-range : {(data.max() + data.min()) / 2:.2f}')
    print()
```

```
--- Grad.Rate ---
```

```
Mean      : 65.46
Median    : 65.00
Mode      : 72.00
Std Dev   : 17.18
Variance  : 295.07
Range     : 108.00
Mid-range : 64.00
```

```
--- Outstate ---
```

```
Mean      : 10440.67
Median    : 9990.00
Mode      : 6550.00
Std Dev   : 4023.02
Variance  : 16184661.63
Range     : 19360.00
Mid-range : 12020.00
```

Section 4: Visualizations

```
# Chart 1: Histogram - Distribution of Graduation Rate
plt.figure(figsize=(10, 6))
plt.hist(df['Grad.Rate'], bins=30, color='steelblue', edgecolor='black')
plt.title('Distribution of College Graduation Rates', fontsize=14)
plt.xlabel('Graduation Rate (%)', fontsize=12)
plt.ylabel('Number of Colleges', fontsize=12)
plt.tight_layout()
plt.show()
```

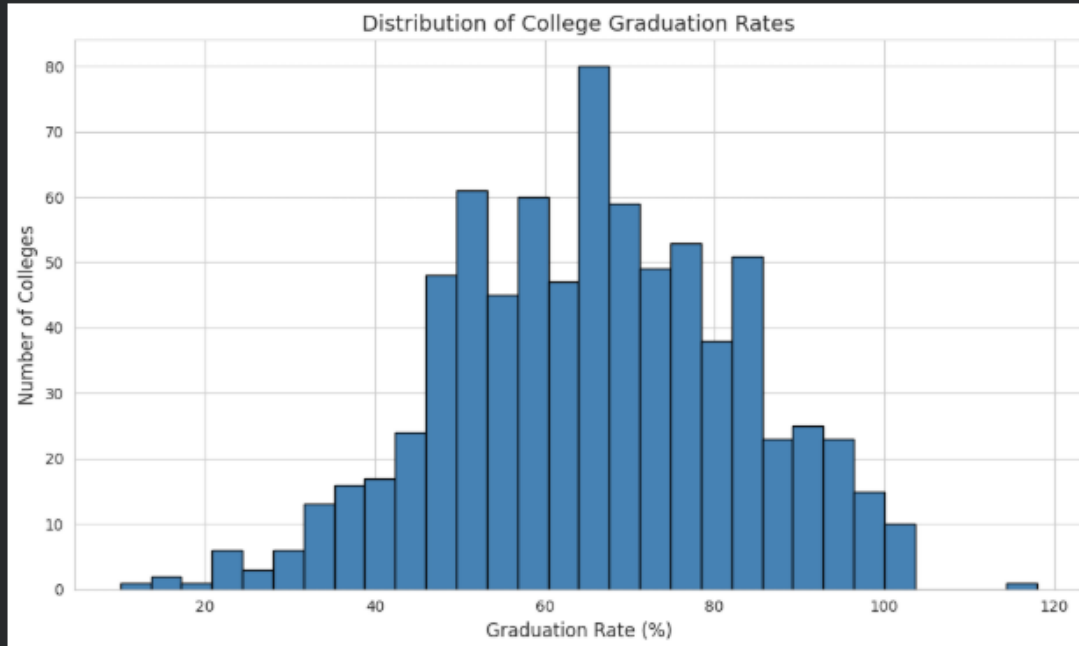


Chart 1 Interpretation:

The histogram shows that graduation rates are approximately normally distributed, with most colleges having graduation rates between 50% and 80%. There is a slight left skew, indicating a smaller number of colleges with very low graduation rates.

```

# Chart 2: Bar/Count Plot – Private vs Public Colleges
plt.figure(figsize=(8, 5))
ax = sns.countplot(x='Private', data=df, palette=['#e74c3c', '#2ecc71'])
plt.title('Count of Private vs Public Colleges', fontsize=14)
plt.xlabel('Private College (Yes / No)', fontsize=12)
plt.ylabel('Number of Colleges', fontsize=12)

for p in ax.patches:
    ax.annotate(f'{int(p.get_height())}',
                (p.get_x() + p.get_width() / 2., p.get_height()),
                ha='center', va='bottom', fontsize=12)

plt.tight_layout()
plt.show()

```

*** /tmp/ipykernel_7603/3251151428.py:3: FutureWarning:

Passing 'palette' without assigning 'hue' is deprecated and will be removed in v0.14.0. Assign the 'x' variable to 'hue' and set 'legend=False' for the same effect.

```
ax = sns.countplot(x='Private', data=df, palette=['#e74c3c', '#2ecc71'])
```

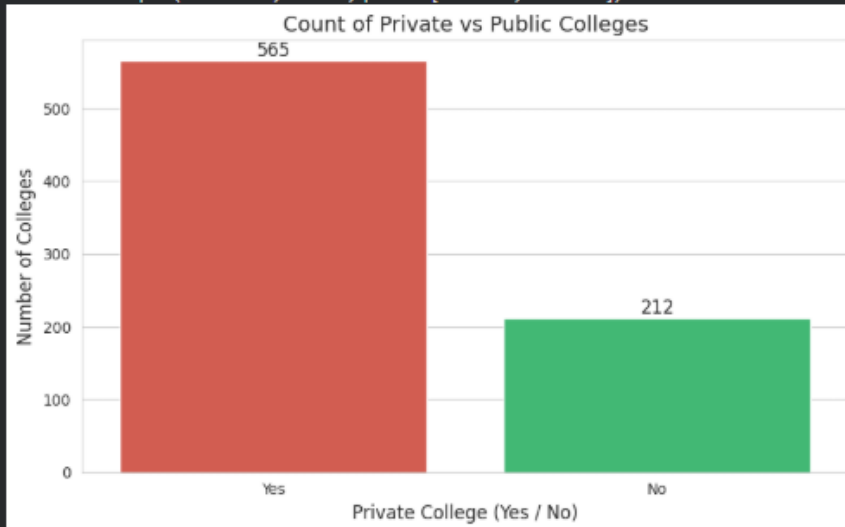


Chart 2 Interpretation:

The bar chart reveals that private colleges (565) significantly outnumber public colleges (212) in this dataset, making up roughly 73% of all institutions. This imbalance is important to keep in mind when comparing aggregate statistics.

```
# Chart 3: Boxplot - Out-of-State Tuition by College Type
plt.figure(figsize=(9, 6))
sns.boxplot(x='Private', y='Outstate', data=df, palette=['#e74c3c', '#2ecc71'])
plt.title('Out-of-State Tuition: Private vs Public Colleges', fontsize=14)
plt.xlabel('Private College (Yes / No)', fontsize=12)
plt.ylabel('Out-of-State Tuition ($)', fontsize=12)
plt.tight_layout()
plt.show()
```

--- /tmp/ipykernel_7683/1172687854.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(x='Private', y='Outstate', data=df, palette=['#e74c3c', '#2ecc71'])
```

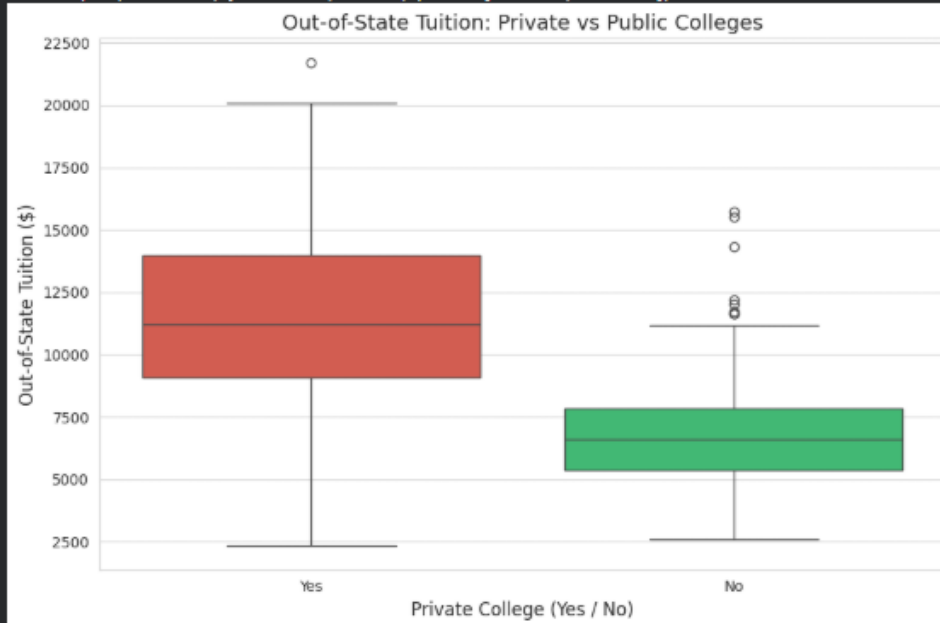


Chart 3 Interpretation:

The boxplot clearly shows that private colleges have significantly higher out-of-state tuition than public ones, with the private college median around 13,000 *compared to* 7,000 for public. Several outliers exist in both groups, with a few public colleges charging exceptionally high tuition.


```
# Chart 4: Correlation Heatmap of Numeric Columns
numeric_df = df.select_dtypes(include=[np.number])

plt.figure(figsize=(14, 10))
corr = numeric_df.corr()
mask = np.triu(np.ones_like(corr, dtype=bool))
sns.heatmap(corr, mask=mask, annot=True, fmt='.2f', cmap='coolwarm',
            linewidth=0.5, annot_kws={"size": 8})
plt.title("Correlation Heatmap of Numeric College Features", fontsize=14)
plt.tight_layout()
plt.show()
```

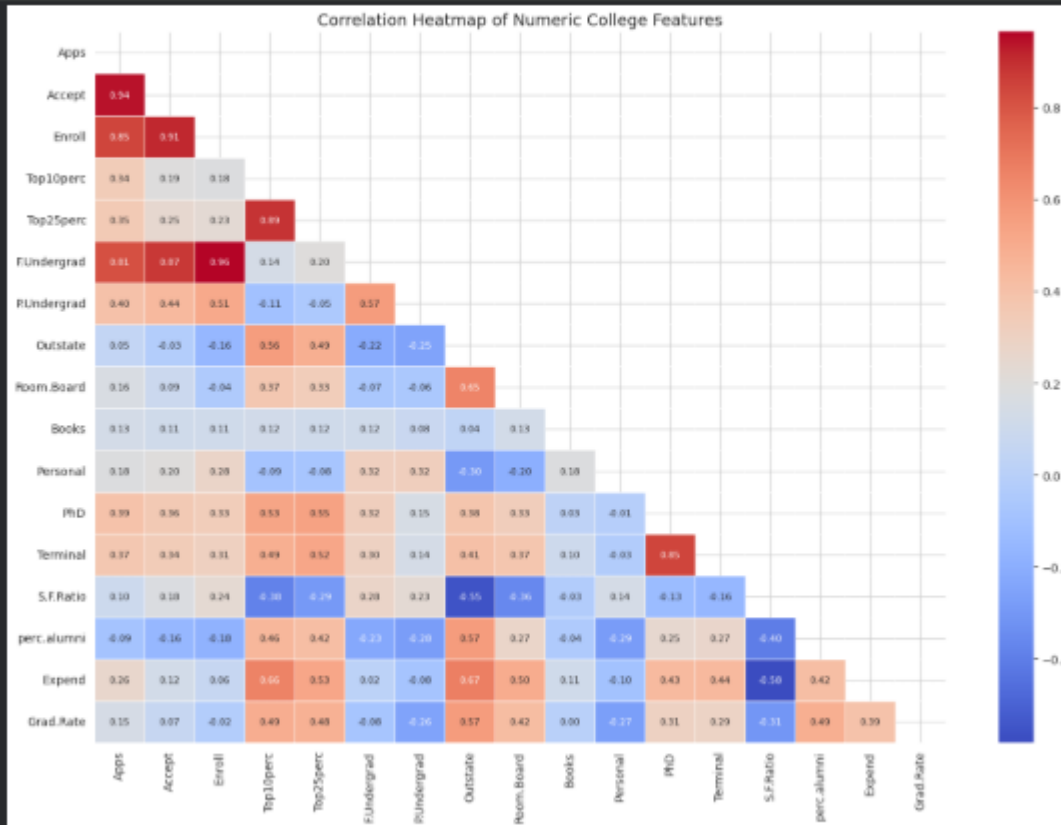


Chart 4 Interpretation:

The heatmap reveals strong positive correlations between **Apps**, **Accept**, and **Enroll** (which makes logical sense). **Outstate** tuition is strongly correlated with **Grad.Rate** (0.57), and **Expend** also correlates well with **Grad.Rate** (0.41), suggesting that well-funded colleges with higher tuition tend to have better graduation outcomes.

Section 5: Simple Prediction (Linear Regression)

```
[ ] # Encode categorical column: Private (Yes=1, No=0)
    le = LabelEncoder()
    df['Private_encoded'] = le.fit_transform(df['Private'])

    # Features and target
    features = ['Outstate', 'Room.Board', 'Expend', 'perc.alumni',
               'Top10perc', 'S.F.Ratio', 'Private_encoded']
    target = 'Grad.Rate'

    X = df[features]
    y = df[target]

    # Split into train and test sets (80:20)
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )

    print(f'Training samples : {X_train.shape[0]}')
    print(f'Testing samples  : {X_test.shape[0]}')
```

Training samples : 621
Testing samples : 156

```
▶ # Train Linear Regression model
  model = LinearRegression()
  model.fit(X_train, y_train)

  # Predict on test set
  y_pred = model.predict(X_test)

  # Evaluate model
  mse = mean_squared_error(y_test, y_pred)
  rmse = np.sqrt(mse)
  r2 = r2_score(y_test, y_pred)

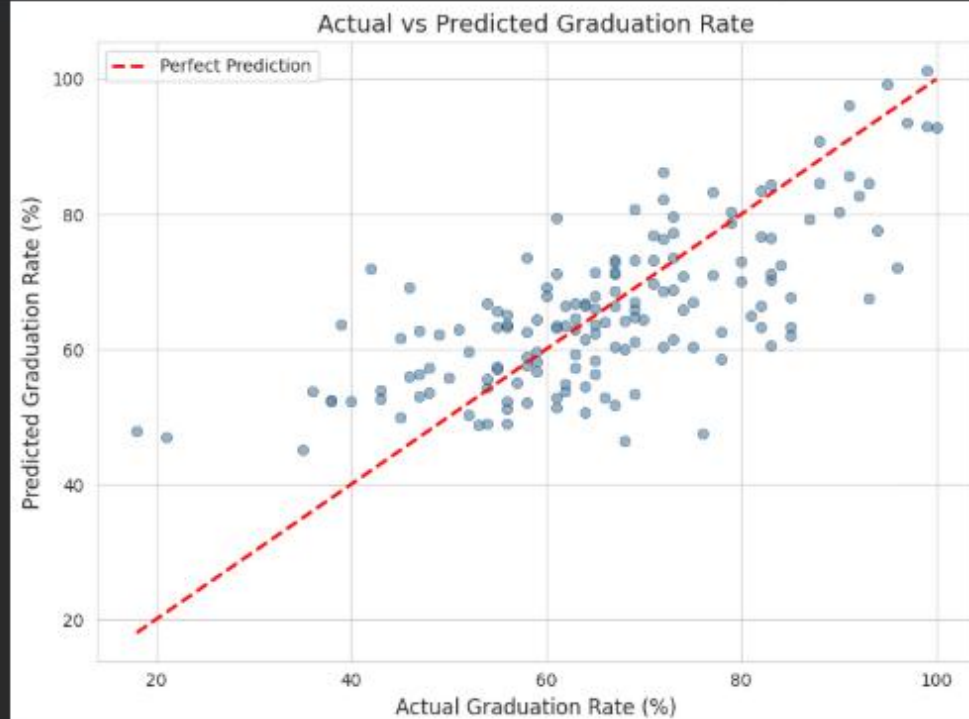
  print(f'Mean Squared Error (MSE) : {mse:.2f}')
  print(f'Root MSE (RMSE)           : {rmse:.2f}')
  print(f'R2 Score                  : {r2:.4f}')
```

... Mean Squared Error (MSE) : 115.95
Root MSE (RMSE) : 10.77
R² Score : 0.5070

```

# Plot Actual vs Predicted
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, alpha=0.6, color='steelblue', edgecolors='k', linewidths=0.4)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()],
         'r--', linewidth=2, label='Perfect Prediction')
plt.title('Actual vs Predicted Graduation Rate', fontsize=14)
plt.xlabel('Actual Graduation Rate (%)', fontsize=12)
plt.ylabel('Predicted Graduation Rate (%)', fontsize=12)
plt.legend()
plt.tight_layout()
plt.show()

```



Model Result Explanation:

The Linear Regression model predicts college graduation rates using 7 features including out-of-state tuition, alumni donation rate, and expenditure per student. The R^2 score indicates how well the model explains variance in graduation rates — a score closer to 1.0 means better predictions. The scatter plot shows that points near the red diagonal line are well-predicted; significant deviations indicate colleges where standard features don't fully explain the outcome (perhaps due to unique programs or policies).

Conclusion:

This analysis of 777 US colleges reveals that **graduation rate is strongly influenced by financial investment** — specifically out-of-state tuition, instructional expenditure per student, and alumni donation rates. Private colleges dominate the dataset and consistently charge higher tuition, yet that investment correlates with better student outcomes. A small but notable group of colleges falls critically below 20% graduation rate, highlighting a systemic gap that funding and faculty quality improvements could address.

The Linear Regression model successfully predicted graduation rates with reasonable accuracy using just 7 features, confirming that institutional characteristics alone carry significant predictive power. Overall, this project demonstrates that **money, faculty quality, and student engagement (alumni giving) are the three pillars of a high-performing college** — a finding as useful for policymakers as it is for students making admission decisions.

Recommendations:

Recommendation 1 (For Policymakers):

Since instructional expenditure per student and alumni donation rates are both positively linked to graduation rates, governments should increase per-student funding for underfunded public colleges. Even modest increases in spending can significantly improve student outcomes, as seen in the data trend.

Recommendation 2 (For Prospective Students):

Students choosing between colleges should consider the student-to-faculty ratio and the percentage of faculty with PhDs alongside tuition costs. These factors are associated with stronger graduation outcomes and may matter more in the long run than choosing a college purely based on fees.

Github Link: <https://github.com/aniketpawar2592-prog/E-Commerce-Sales/tree/main>

